KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

NAME:

TEACHER:

MATHEMATICS

Year 9 5.2

Yearly Exam

2026

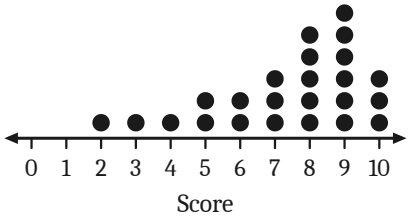
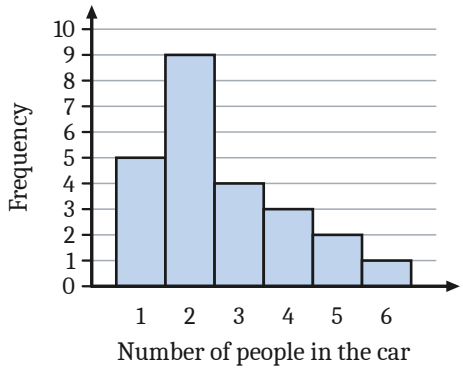
Instructions

1. Reading time: 5 minutes
2. Working time: 90 minutes
3. Attempt all questions
4. Show ALL working where needed
5. A calculator is permitted

Sections and Marks

Section 1 - Data	/25
Section 2 - Surface area and volume	/24
Section 3 - Mixed Topics	/21
Total marks	**/70**

Section 1: Data

	Multiple Choice:	Mark
1	Which of the following is **numerical continuous** data? A. The number of goals in a hockey match B. The colours of cars in a car park C. Shirt sizes (S, M, L, XL) D. The time taken to swim 50 m	1
2	The dot plot shows the scores (out of 10) of 24 students in a spelling quiz. Which statement best describes the data? A. Positively skewed, clustered at 8 and 9 B. Negatively skewed, clustered at 8 and 9 C. Negatively skewed, clustered at 2 and 3 D. Symmetrical, clustered at 8 and 9 	1
3	The stem-and-leaf plot shows the ages of the 15 people in a yoga class. What is the median age? A. 30 B. 31 C. 32 D. 36 Stem | Leaf 1 | 8 9 2 | 3 5 5 7 3 | 0 2 4 6 6 6 4 | 1 5 5 | 8 Key: 1 | 8 = 18 years	1
4	The histogram shows the number of people in each of 24 cars that arrived at a beach car park. What is the mode? A. 2 B. 3.5 C. 6 D. 9 	1
5	A water polo team scored these numbers of goals in 10 games: 5, 7, 6, 8, 6, 21, 7, 5, 6, 8 Which measure of centre changes the most if the outlier, 21, is removed? A. The mode B. The median C. The mean D. All three change by the same amount	1

Section 1: Data (continued)

Short Answer:

<table>
<tr><th></th><th>Question</th><th>Working Out</th><th>Mark</th></tr>
<tr><td></td><td colspan="2">Use this set of data for questions 6 to 9.
The numbers of laps swum by 10 students at swimming training were:
9, 15, 11, 12, 18, 12, 14, 10, 16, 13</td><td></td></tr>
<tr><td>6</td><td>Find the mean.
Note: Mean = sum of the scores ÷ number of scores</td><td></td><td>1</td></tr>
<tr><td>7</td><td>Find the median.</td><td></td><td>1</td></tr>
<tr><td>8</td><td>Find the mode.</td><td></td><td>1</td></tr>
<tr><td>9</td><td>Find the range.
Note: Range = highest score − lowest score</td><td></td><td>1</td></tr>
<tr><td>10</td><td>Students were asked how many times they went to the beach last week.
<table>
<tr><th>Visits (x)</th><th>Frequency (f)</th><th>f × x</th></tr>
<tr><td>0</td><td>2</td><td></td></tr>
<tr><td>1</td><td>6</td><td></td></tr>
<tr><td>2</td><td>8</td><td></td></tr>
<tr><td>3</td><td>5</td><td></td></tr>
<tr><td>4</td><td>3</td><td></td></tr>
<tr><td>5</td><td>2</td><td></td></tr>
<tr><td>Total</td><td></td><td></td></tr>
</table>
a. Complete the table.
b. Find the mean number of visits, correct to 2 decimal places.
c. Find the median number of visits.
d. Find the range.
Note: Mean = total of f × x ÷ total frequency
Range = highest score − lowest score</td><td>a.

b.

c.

d.</td><td>1

1

1

1</td></tr>
</table>

Section 1: Data (continued)

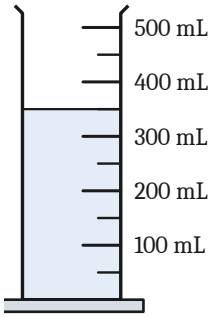
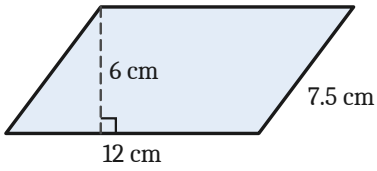
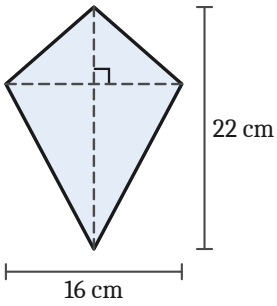
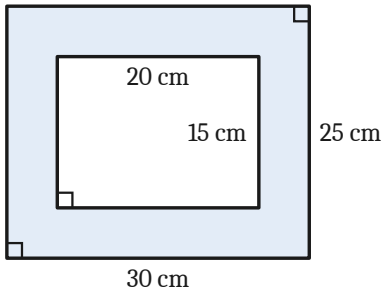
	Question	Working Out	Mark
11	The delivery times (in minutes) of 10 orders from each of two pizza shops are shown below. **Pete's Pizza**: 18, 25, 32, 21, 27, 35, 29, 24, 41, 26 **Pizza Point**: 15, 22, 19, 28, 17, 23, 16, 30, 21, 18		
a	Draw an ordered back-to-back stem-and-leaf plot of the data. Include a key.		2
b	Find the median delivery time for each shop.		1
c	Which shop delivers faster? Give a reason.		1

	Section 1: Data (continued)		
	Worded Problems:		
	Question	**Working Out**	**Mark**
12	A surf shop has 10 employees, including the manager. Their weekly wages are: $860, $920, $780, $1050, $840, $3600, $900, $880, $760, $950 Note: Mean = sum of the scores ÷ number of scores		
a	Find the mean weekly wage.		1
b	Find the median weekly wage.		1
c	The manager earns $3600. Find the mean weekly wage of the other 9 employees, correct to the nearest cent.		1
d	Which measure, the mean or the median, better describes the typical weekly wage at this shop? Give a reason.		1

Section 1: Data (continued)

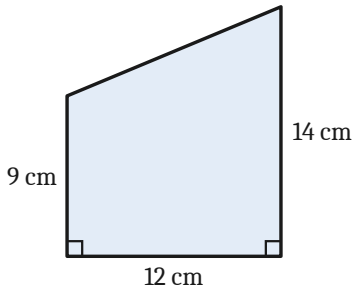
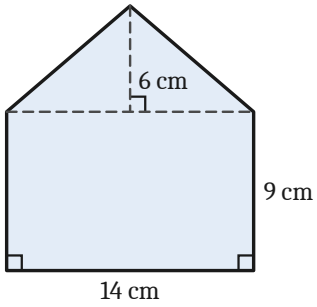
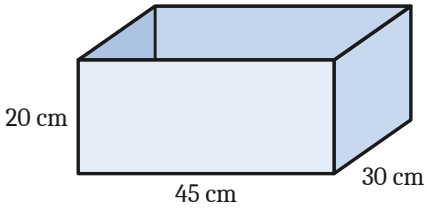
	Question	Working Out	Mark
13	The Kestrels and the Tigers are two netball teams. The back-to-back stem-and-leaf plot shows the number of goals each team scored in its 12 games this season. Note: Mean = sum of the scores ÷ number of scores Note: Range = highest score − lowest score	Kestrels | Stem | Tigers 2 | 2 7 8 8 7 5 4 1 | 3 | 3 6 9 6 4 3 2 1 0 | 4 | 1 4 5 7 5 | 0 3 7 Key: 4 | 1 = 41 goals	
a	Find the mean number of goals for each team, correct to 1 decimal place.		1
b	Find the median number of goals for each team.		1
c	Which team usually scores more goals? Give a reason.		1
d	Which team is more consistent? Use the range of each team to explain.		1
	Total		**/25**

Section 2: Surface area and volume

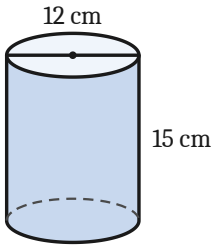
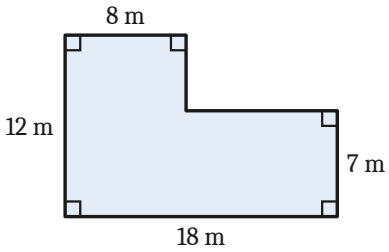
	Multiple Choice:	Mark
1	A rainwater tank holds 4.6 kL of water. How many litres is this? A. 460 L B. 46 000 L C. 4600 L D. 0.0046 L	1
2	The scale on this measuring cylinder is shown. What is the absolute error of a reading taken with this measuring cylinder? Note: absolute error = ½ × precision (the smallest marked unit on the scale) A. ±25 mL B. ±50 mL C. ±100 mL D. ±0.5 mL 	1
3	What is the area of this parallelogram? Note: Area of a parallelogram: $A = bh$ A. 90 cm^2 B. 36 cm^2 C. 39 cm^2 D. 72 cm^2 	1
4	What is the area of this kite? Note: Area of a kite: $A = ½xy$ (x and y are the diagonals) A. 352 cm^2 B. 176 cm^2 C. 88 cm^2 D. 38 cm^2 	1
5	A photo frame is a 30 cm by 25 cm rectangle with a 20 cm by 15 cm hole for the photo. What is the area of the frame (the shaded part)? Note: Area of a rectangle: $A = lb$ A. 750 cm^2 B. 1050 cm^2 C. 450 cm^2 D. 100 cm^2 	1

Section 2: Surface area and volume (continued)

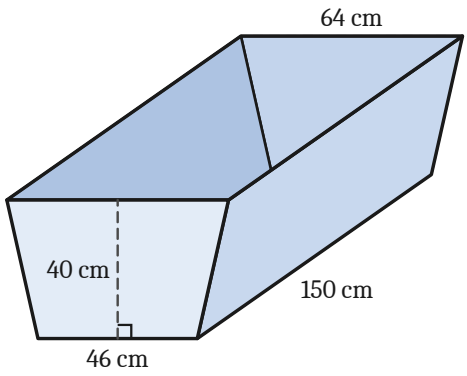
Short Answer:

	Question	Working Out	Mark
6	Find the perimeter of this shape.  Note: Pythagoras' theorem: $c^2 = a^2 + b^2$		2
7	Find the area of this shape.  Note: Area of a rectangle: $A = lb$ Note: Area of a triangle: $A = \frac{1}{2}bh$		3
8	A storage tub is a rectangular prism with **no lid** (it is open at the top). Find the surface area of the tub.  Note: Surface area = the total area of the faces; area of a rectangle: $A = lb$		2

Section 2: Surface area and volume (continued)

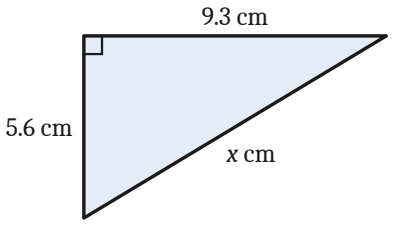
	Question	Working Out	Mark
9	This closed cylinder has a diameter of 12 cm and a height of 15 cm.  a. Find its volume, correct to 1 decimal place. b. Find its surface area, correct to 1 decimal place. Note: Volume: $V = \pi r^2 h$ Note: Surface area: $SA = 2\pi r^2 + 2\pi rh$	a. b.	2 2
	Worded Problems:		
	Question	**Working Out**	**Mark**
10	The plan shows an L-shaped lawn at a primary school. All of the corners are right angles. Note: Area of a rectangle: $A = lb$  		
a	Garden edging is laid around the outside of the lawn. How many metres of edging are needed?		1
b	Find the area of the lawn.		2
c	New turf costs $12.40 per square metre. Find the cost of the turf for the whole lawn.		1

Section 2: Surface area and volume (continued)

	Question	Working Out	Mark
11	A farmer's water trough is a prism 150 cm long, open at the top. Its end is a trapezium with the measurements shown. 		
a	Find the area of the trapezium-shaped end of the trough. Note: Trapezium: $A = \frac{1}{2}h(a + b)$		2
b	Find the volume of the trough in cm^3. Note: Volume of a prism: $V = A \times l$ (A = area of the end, l = length of the prism)		1
c	How many litres of water does the trough hold when it is full? Note: 1 L = 1000 cm^3		1
	Total		**/24**

Section 3: Mixed Topics (Chapters 1–8)

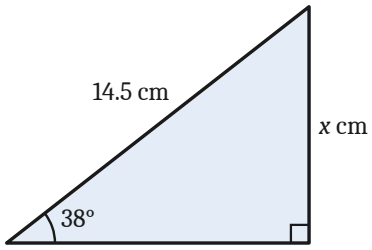
Multiple Choice:

		Mark
1	If $a = 5$ and $b = 3$, evaluate $2a^2 - b$. A. 7 B. 17 C. 47 D. 97	1
2	Find the value of x. Select the closest answer. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ A. 14.9 B. 10.9 C. 7.4 D. 3.7 	1
3	A skateboard is marked at $145. In a sale it is discounted by 15%. What is the selling price? A. $166.75 B. $130.00 C. $123.25 D. $21.75	1
4	Simplify $4m^6 \times 3m^2$. A. $12m^8$ B. $7m^8$ C. $12m^{12}$ D. $12m^4$	1
5	Ella is a nurse. She earns an annual salary of $81 640. How much is she paid per fortnight? Note: 1 year = 52 weeks = 26 fortnights = 12 months A. $1570.00 B. $3140.00 C. $3401.67 D. $6803.33	1

Short Answer:

	Question	Working Out	Mark
6	Expand and simplify: $3(4x - 5) - 2(x - 6)$		2
7	Liam invests $7250 for 3 years at 4.2% p.a. simple interest. Calculate the interest he earns. Note: Simple interest: $I = Prn$		2

Section 3: Mixed Topics (Chapters 1–8) (continued)

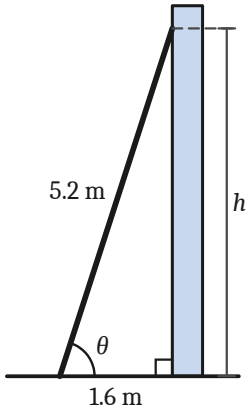
	Question	Working Out	Mark
8	Find the value of x, correct to 2 decimal places.  Note: $\sin\theta = \frac{\text{opp}}{\text{hyp}}$, $\cos\theta = \frac{\text{adj}}{\text{hyp}}$, $\tan\theta = \frac{\text{opp}}{\text{adj}}$		2
9	a. Write 0.000 724 in scientific notation. b. Evaluate $(4.8 \times 10^{6}) \times (3.5 \times 10^{-2})$, giving your answer in scientific notation.	a. b.	1 1
10	Solve each equation. a. $5x - 8 = 27$ b. $7m - 2(m + 3) = 3m + 10$	a. b.	1 1

Section 3: Mixed Topics (Chapters 1–8) (continued)

Worded Problems:

	Question	Working Out	Mark
11	Mia works in a bakery. Her normal rate of pay is $26.80 per hour, and she is paid according to the award shown below. Last week Mia worked 44 hours. **Pay rate** / **Hours worked** Normal rate / For 0 to 38 hours Time-and-a-half / For the next 3 hours Double time / For each hour after that		
a	Calculate Mia's pay for the first 38 hours.		1
b	Calculate Mia's total pay for the 44 hours.		2

Section 3: Mixed Topics (Chapters 1–8) (continued)

	Question	Working Out	Mark
12	A ladder 5.2 m long leans against a vertical wall. The foot of the ladder is 1.6 m from the wall, on level ground. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ Note: $\sin\theta = \frac{\text{opp}}{\text{hyp}}$, $\cos\theta = \frac{\text{adj}}{\text{hyp}}$, $\tan\theta = \frac{\text{opp}}{\text{adj}}$ 		
a	Find h, the height the ladder reaches up the wall, correct to 2 decimal places.		1
b	Find the angle θ that the ladder makes with the ground, correct to the nearest degree.		1
c	For safety, a ladder should make an angle of 75° with the ground. A longer ladder is placed at 75° so that its top reaches 6.5 m up the wall. How long is this ladder? Answer correct to 2 decimal places.		1
	Total		**/21**